# MILITARY SPECIFICATION

# SNOWPLOW, RAILWAY, PUSH, WEDGE TYPE,

# LOCOMOTIVE PILOT MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers a commercial railway wedge-type pilot snowplow for clearing snow from track to permit uninterrupted locomotive operation on railways within the continental limits of the United States.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-C-650 | - Crates, Wood, Open and Covered. |

Military

| | |
|---|---|
| MIL-P-116 | - Preservation-Packaging, Methods of. |
| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| MIL-P-3321 | - Painting and Marking: Railway Motive Power and Work Equipment. |
| MIL-B-26195 | - Boxes, Wood-Cleated, Skidded, Load-Bearing Base. |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-129 | - Marking for Shipment and Storage. |

FSC 2230

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|---|---|
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-1186 | - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods. |

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Manual of Standards and Recommended Practices:
Mechancial Division:
Section A - Specifications for Materials.
M-110 - Rivet Steel and Rivets.
M-116 - Steel, Structural Shapes, Plates and Bars.
M-117 - Heavy and Light Gage Carbon Steel Sheets and Strip Steel.
M-125 - Machine Bolts, and nuts.
M-201 - Steel castings.

Specifications for Design, Fabrication and Construction of Freight Cars:
Equipment Diagram, AAR Plate B, for Unlimited Interchange Service.

Electrical Manual of Standards and Recommended Practice Section 6, Welding, and Cutting.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)



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AMERICAN WELDING SOCIETY, INC.

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N. W. Seventh Street, Miami, FL 33125.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

## 3. REQUIREMENTS

3.1 Description. The snowplow shall be of the center wedge design, nonadjustable, and shall attach to the front or rear of a 56-1/2-inch gage diesel electric locomotive ranging in sizes from 25 to 70 tons. The plow for the various size locomotives shall be of the same design, but the arrangement for bolting shall be suitable for the particular model and make of locomotive specified by the contracting officer (see 6.2). The snowplow shall be a commercial model of the standard product of the supplier and shall have been manufactured and marketed for at least 1 year prior to the date of the opening of the technical proposal. Normal product improvements are permitted.

3.2 First article.

3.2.1 First-produced snowplow. The supplier shall furnish a snowplow for examination and test within the time frame specified to prove that his production methods will produce snowplows that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.2 and 6.3).

3.3 Materials. Unless otherwise specified herein, materials and components shall conform to applicable AAR and commerical specifications currently used by the railroad industry for this type of equipment.



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3.3.1 Steel. Unless otherwise specified, structural shapes, plates, and bars shall conform to Grade B, and cold pressed plates to Grade C, of AAR M-116, and sheets and strip steel to AAR M-117. Castings shall conform to AAR M-201, Grade A, unannealed.

3.3.2 Bolts, nuts, studs, and rivets. Material for bolts and studs shall conform to AAR-M-125. Rivet material shall conform to AAR-M-110.

3.3.3 Substitution of materials. Materials not conforming to the specifications and drawings referenced herein shall not be substituted unless such substitutions are specifically approved by the contracting officer. Substitutions will be approved only if it is clearly demonstrated that an improvement in operating characteristics, a saving in weight, conservation of critical or strategic materials, or a reduction in cost without sacrifice of reliability can be accomplished thereby, and that such substitutions will not preclude the subsequent use of specified materials in effecting repairs or replacements. Request for approval of such substitutions shall be accompanied by completely detailed supporting data which clearly illustrate the use of the proposed substitute materials.

3.4 Interchangeability of parts. Components shall be interchangeable (without requiring modification for replacement) with similar components furnished under the same contract or order.

3.5 Threaded parts. Threaded parts shall conform to Handbook H28. The American National fine thread series shall be used for threaded parts less than 1/4-inch diameter. The Unified or American National coarse thread series shall be used for threaded parts 1/4-inch diameter and larger; however, the American National fine thread series may be employed for these sizes when applicable. Pipe threads shall conform to American National taper pipe thread series.

3.6 Welding. The surface of all parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Unless otherwise specified, or shown on the drawings, welds shall develop the strength required for the parts connected. All welding shall be in accordance with the latest applicable codes of the American Welding Society as contained in the Welding Handbook. Welders shall be qualified in accordance with the applicable tests specified in the Welding Handbook, or with the Qualification Test for Welding Operators as contained in Section 6 of the Electrical Section of the Manual of Standard and Recommended Practice.



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3.7 Steel fabrication. Steel used in the fabrication of the snowplow and equipment shall be free from kinks and sharp bends, straightening shall be done only by methods that will not cause damage to the metal. Shearing, cutting, chipping, and similar operations shall be done neatly and accurately and all corners shall be neat and true. Flame cutting may be employed. Burned surfaces of flame-cut material shall be ground or machined to remove slag, flame gougings, and cooling checks. All re-entrant cuts shall be made in the best possible manner. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Care shall be taken to avoid overheating and heated metal shall be allowed to cool slowly.

3.8 Tolerances and fits. Tolerances and fits shall conform to all limitations specified herein, and when not specified, shall conform to standard commercial practice.

3.9 Maintainability. The plow shall require only such maintenance which can be accomplished with handtools while the plow is in service.

3.10 Dimensions and weights. General dimensions and weights of the snowplow shall be as follows:

| | |
|---|---|
| WIDTH | 10 FEET |
| HEIGHT (at nose) | 18 INCHES |
| HEIGHT (at rear) | 44 INCHES |
| WEIGHT | 1,500 POUNDS |

(Above dimensions are for guidance only)

3.10.1 Clearance. Clearance dimensions shall not exceed those shown on AAR Plate B.

3.11 Design and construction.

3.11.1 Design. The snowplow shall be designed to move snow from the center of the track to both sides, and shall be bolted to the locomotive with no provisions for changing the height or angularity of the unit. The snowplow shall be installed below the coupler so that it in no way interferes with the coupling of the end of the locomotive to a car or train. The steps, coupler pin, lifting lever, and hose couplings shall be accessible for use by the brakeman. All hardware for mounting the snowplow to the locomotive shall be supplied with each unit. The snowplow shall maintain a distance from the top of the rail to the bottom of the plow at a minimum 3 inches.



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3.11.2 Construction. The snowplow shall be either constructed of cast steel with all the supports and bracing required for fastening the plow to the locomotive cast integral with the plow or fabricated from plate steel, and bolted, riveted, or welded together with the necessary supports and bracing. The face and flare of the plow shall be at least 1/2-inch thick and radially contoured with a plowing angle which will clear the track of snow to operate trains.

3.12 Identification marking. The snowplow shall be identified in accordance with MIL-STD-130. Identification plates shall conform to MIL-P-514.

3.13 Treatment and painting. The portions of the snowplow and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-P-3321.

3.14 Workmanship. All parts, components and assemblies of the snowplow, including castings, forgings, molded parts, stampings, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, pits, scale, flux, and other extraneous material. All edges shall be rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, inspected, and tested in accordance with applicable AAR specifications as modified herein.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) First-produced snowplow inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).



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4.3 First-produced snowplow inspection. The first-produced snowplow which shall have been produced from production tooling shall be examined and tested to determine conformance to the requirements of this specification. The first-produced snowplow shall be compared with the requirements of this specification and the suppliers published literature and calculations to verify conformance to this specification.

4.3.1 Examination. The first-produced snowplow shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced snowplow shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each snowplow shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each snowplow shall be tested as specified in 4.5.2.1. Failure of any test shall be cause for rejection.

4.5 Inspection procedures.

4.5.1 Examination. Each snowplow shall be examined for the following defects:

101. Parts or components missing.
102. Parts or components do not function.
103. Assembly not complete or not correct.
104. Material not as specified.
105. Screw threads not as specified.
106. Dimensions and weight not as specified.
107. Level not as specified.
108. Identification markings or instruction plates missing.
109. Treatment or painting not as specified.

4.5.2 Tests.

4.5.2.1 Installation. The snowplow shall be installed or mounted on a locomotive of the make and model specified for all tests (see 6.2). Inability to mount snowplow to locomotive specified or clearance not as specified in 3.11.1 or 3.12 shall constitute failure of this test.

4.5.2.2 Coupling test. With snowplow installed on a locomotive equipped with an AAR type "E" coupler, connect locomotive to a car at the same end the snowplow is mounted. Inability to connect locomotive



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to car, inability to use steps on locomotive, inacessibility of couple pin lifting lever, or hose couplings shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection a unit of product is considered a completed snowplow prepared for shipment with all components attached to snowplow.

4.6.1.2 Examination. Each snowplow shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

110. Materials, methods, and containers not as specified. Each incorrect material, method, or container shall constitute one defect.
111. Disassembly not as specified.
112. Blocking, bracing, and anchoring not as specified.
113. Marking illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY.

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be Level A or C, as specified (see 6.2).

5.1.1 Level A. Methods, materials, and scope of cleaning, drying, preservation, including unit protection and packaging, shall be in accordance with MIL-P-116, and as specified herein.

5.1.1.1 Disassembly. Disassembly of the snowplow shall be the minimum necessary to safeguard parts vulnerable to damage and loss and to accomplish reduction in cube, in accordance with the manufacturer's recommendation. Removed bolts, nuts, screws, washers, and pins shall be reinstalled in mating parts to prevent their loss. Disassembled parts shall be clearly and legibly marked to show identification and proper location on assembled items. A part shall not be removed unless it can be reassembled in the field without special tools.

5.1.1.2 Preservatives. Preservative specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.



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5.1.1.3 Unpainted metal surfaces. Unpainted exterior metal surfaces not specified herein requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative.

5.1.2 Level C. Each complete snowplow shall be preserved and packaged to afford protection against deterioration and damage during shipment from the supplier to initial destination.

5.2 Packing. Packing shall be Level A or C as specified (see 6.2).

5.2.1 Level A. Each complete snowplow shall be packed in a wood box conforming to MIL-B-26195, Type II, Class 2 style optional, with rubbing strips or in a crate conforming to PPP-C-650, Style A, Type V, Class 1 with a shroud as specified in the appendix to the crate specification. Blocking, bracing, and anchoring of the contents shall be in accordance with MIL-STD-1186.

5.2.2 Level B. Packing shall be the same as for Level A in 5.2.1 except boxes shall be Type I and crates shall be Type V, Style A or B, open.

5.2.3 Level C. Each complete snowplow shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings. Containers shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The snowplow is intended to be used on a locomotive during normal rail operations to clear light snow from the tracks.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Make and model of locomotive (see 3.1).
(c) Time frame required for submission of first-produced model (see 3.2.1).
(d) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).



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6.3 First-produced snowplows. Any changes or deviations of production snowplows from the approved first-produced snowplows during production will be subject to the approval of the contracting officer. Approval of the first-produced snowplow will not relieve the supplier of his obligation to furnish snowplows conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts list, and suppliers maintenance and operation manual to be furnished with each snowplow.

6.5 Provisions. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of snowplows. A suggested paragraph is as follows:

"Shipment of snowplows shall include repair parts maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.6 Government serial numbers. The contracting officer should furnish the Government serial numbers to the supplier, upon award of the contract or order for the snowplow.

6.7 Suppliers published literature. The contracting officer will determine from a review of literature submitted by the bidders that the plows offered have been used sucessfully by American Railroads (see 3.1).

Custodian:

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Preparing activity:

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DOCUMENT IDENTIFIER AND TITLE MIL-S-12115A(ME) Snowplow, Railway, Push, Wedge Type, Locomotive Pilot Mounted

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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